Experiment 3: Pharmacy Database Using MongoDB

Aim: To create a pharmacy database with schema validation, insert sample data, and perform multi-collection aggregation queries.

a) Switch to database

```
use pharmacyDB
```

b) Create collections with validation

```
db.createCollection("customers", { validator: { $jsonSchema: { bsonType:
"object", required: ["custId","name","age","phone"], properties: { custId:
{ bsonType: "int" }, name: { bsonType: "string" }, age: { bsonType: "int",
minimum: 18 }, phone: { bsonType: "string" } } } } });

db.createCollection("medicines", { validator: { $jsonSchema: { bsonType:
"object", required:
["medId","medName","supplierId","price","stock","expiryDate"], properties:
{ medId: { bsonType: "int" }, medName: { bsonType: "string" }, supplierId:
{ bsonType: "int" }, price: { bsonType: "int" }, stock: { bsonType: "int",
minimum: 0 }, expiryDate: { bsonType: "date" } } } } });

db.createCollection("suppliers", { validator: { $jsonSchema: { bsonType:
"object", required: ["supplierId","name","city"], properties: { supplierId:
{ bsonType: "int" }, name: { bsonType: "string" }, city: { bsonType:
"string" } } } } });

db.createCollection("orders", { validator: { $jsonSchema: { bsonType:
"object", required: ["orderId","custId","medId","quantity","orderDate"],
properties: { orderId: { bsonType: "int" }, custId: { bsonType: "int" },
medId: { bsonType: "int" }, quantity: { bsonType: "int", minimum: 1 },
orderDate: { bsonType: "date" } } } } });

db.createCollection("pharmacists", { validator: { $jsonSchema: { bsonType:
"object", required: ["pharmaId","name","joiningDate"], properties: {
pharmaId: { bsonType: "int" }, name: { bsonType: "string" }, joiningDate: {
bsonType: "date" } } } } });
```

c) Insert sample data (10 documents each)

Customers

```
db.customers.insertMany([
 {custId:1,name:"Arjun",age:25,phone:"9876543210"},
 {custId:2,name:"Priya",age:30,phone:"9876501234"},
 {custId:3,name:"Ravi",age:22,phone:"9876512345"},
 {custId:4,name:"Meena",age:28,phone:"9876523456"},
 {custId:5,name:"Amit",age:35,phone:"9876534567"},
 {custId:6,name:"Kiran",age:40,phone:"9876545678"},
 {custId:7,name:"Sita",age:29,phone:"9876556789"},
 {custId:8,name:"Raj",age:33,phone:"9876567890"},
 {custId:9,name:"Anita",age:26,phone:"9876578901"},
 {custId:10,name:"Vikram",age:38,phone:"9876589012"}
]);
```

Suppliers

```
db.suppliers.insertMany([
 {supplierId:1,name:"MedLife Pvt Ltd",city:"Bangalore"},
 {supplierId:2,name:"PharmaCare",city:"Chennai"},
 {supplierId:3,name:"MediWorld",city:"Delhi"},
 {supplierId:4,name:"HealthPlus",city:"Hyderabad"},
 {supplierId:5,name:"LifeMeds",city:"Mumbai"},
 {supplierId:6,name:"BioPharma",city:"Kolkata"},
 {supplierId:7,name:"CareWell",city:"Bangalore"},
 {supplierId:8,name:"Wellness Drugs",city:"Pune"},
 {supplierId:9,name:"MediCure",city:"Ahmedabad"},
 {supplierId:10,name:"GoodHealth Pharma",city:"Jaipur"}
]);
```

Medicines

```
db.medicines.insertMany([

{medId:1,medName:"Paracetamol",supplierId:1,price:20,stock:100,expiryDate:I
SODate("2025-12-31")},

{medId:2,medName:"Amoxicillin",supplierId:2,price:50,stock:80,expiryDate:IS
ODate("2025-06-30")},

{medId:3,medName:"Ibuprofen",supplierId:3,price:30,stock:60,expiryDate:ISOD
ate("2025-11-15")},

{medId:4,medName:"Cetirizine",supplierId:4,price:15,stock:120,expiryDate:IS
ODate("2025-09-10")},

{medId:5,medName:"Azithromycin",supplierId:5,price:60,stock:50,expiryDate:I
SODate("2025-08-25")},
 {medId:6,medName:"Vitamin
C",supplierId:6,price:10,stock:200,expiryDate:ISODate("2026-01-20")},
 {medId:7,medName:"Dolo
650",supplierId:7,price:25,stock:150,expiryDate:ISODate("2025-07-15")},

{medId:8,medName:"Metformin",supplierId:8,price:40,stock:90,expiryDate:ISOD
ate("2025-10-05")},

{medId:9,medName:"Atorvastatin",supplierId:9,price:35,stock:70,expiryDate:I
SODate("2025-11-30")},

{medId:10,medName:"Omeprazole",supplierId:10,price:45,stock:65,expiryDate:I
SODate("2025-12-10")}
]);
```

Pharmacists

```
db.pharmacists.insertMany([
 {pharmaId:1,name:"Dr. Kumar",joiningDate:ISODate("2018-06-10")},
 {pharmaId:2,name:"Dr. Meera",joiningDate:ISODate("2017-07-15")},
 {pharmaId:3,name:"Dr. Ramesh",joiningDate:ISODate("2019-03-20")},
 {pharmaId:4,name:"Dr. Anita",joiningDate:ISODate("2016-08-25")},
 {pharmaId:5,name:"Dr. Vijay",joiningDate:ISODate("2020-01-12")},
 {pharmaId:6,name:"Dr. Sunita",joiningDate:ISODate("2021-09-05")},
 {pharmaId:7,name:"Dr. Karthik",joiningDate:ISODate("2015-11-11")},
 {pharmaId:8,name:"Dr. Rekha",joiningDate:ISODate("2019-02-14")},
 {pharmaId:9,name:"Dr. Arvind",joiningDate:ISODate("2020-05-18")},
 {pharmaId:10,name:"Dr. Nisha",joiningDate:ISODate("2022-04-21")}
]);
```

Orders

```
db.orders.insertMany([
 {orderId:1,custId:1,medId:1,quantity:2,orderDate:ISODate("2025-08-01")},
 {orderId:2,custId:1,medId:2,quantity:1,orderDate:ISODate("2025-08-02")},
 {orderId:3,custId:2,medId:3,quantity:3,orderDate:ISODate("2025-08-03")},
 {orderId:4,custId:3,medId:4,quantity:1,orderDate:ISODate("2025-08-04")},
 {orderId:5,custId:4,medId:5,quantity:2,orderDate:ISODate("2025-08-05")},
 {orderId:6,custId:5,medId:6,quantity:4,orderDate:ISODate("2025-08-06")},
 {orderId:7,custId:6,medId:7,quantity:1,orderDate:ISODate("2025-08-07")},
 {orderId:8,custId:7,medId:8,quantity:2,orderDate:ISODate("2025-08-08")},
 {orderId:9,custId:8,medId:9,quantity:3,orderDate:ISODate("2025-08-09")},
 {orderId:10,custId:9,medId:10,quantity:1,orderDate:ISODate("2025-08-10")}
]);
```

d) Aggregation queries

i) List all orders with customer name and medicine name

```
db.orders.aggregate([
 { $lookup: { from:"customers", localField:"custId", foreignField:"custId",
as:"customerInfo" } },
 { $unwind:"$customerInfo" },
 { $lookup: { from:"medicines", localField:"medId", foreignField:"medId",
as:"medicineInfo" } },
 { $unwind:"$medicineInfo" },
 { $project: { _id:0, orderId:1, customer:"$customerInfo.name",
medicine:"$medicineInfo.medName", quantity:1,
pricePerUnit:"$medicineInfo.price" } }
]);
```

ii) Total sales per medicine

```
db.orders.aggregate([
 { $lookup: { from:"medicines", localField:"medId", foreignField:"medId",
as:"medInfo" } },
 { $unwind:"$medInfo" },
 { $group: { _id:"$medInfo.medName", totalSold:{ $sum:"$quantity" },
totalRevenue:{ $sum:{ $multiply:["$quantity","$medInfo.price"] } } } },
 { $sort: { totalRevenue:-1 } }
]);
```

iii) Supplier-wise revenue

```
db.orders.aggregate([
 { $lookup: { from:"medicines", localField:"medId", foreignField:"medId",
as:"medInfo" } },
 { $unwind:"$medInfo" },
 { $lookup: { from:"suppliers", localField:"medInfo.supplierId",
foreignField:"supplierId", as:"supplierInfo" } },
 { $unwind:"$supplierInfo" },
 { $group: { _id:"$supplierInfo.name", totalRevenue:{ $sum:{
$multiply:["$quantity","$medInfo.price"] } } } },
 { $sort: { totalRevenue:-1 } }
]);
```

iv) Medicines below stock threshold (less than 70 units)

```
db.medicines.aggregate([
 { $match: { stock:{ $lt:70 } } },
 { $lookup: { from:"suppliers", localField:"supplierId",
foreignField:"supplierId", as:"supplierInfo" } },
 { $unwind:"$supplierInfo" },
 { $project: { _id:0, medName:1, stock:1, supplier:"$supplierInfo.name" } }
]);
```

v) Customer order summary

```
db.customers.aggregate([
 { $lookup: { from:"orders", localField:"custId", foreignField:"custId",
as:"ordersList" } },
 { $project: { _id:0, name:1, totalOrders:{ $size:"$ordersList" } } },
 { $sort: { totalOrders:-1 } }
]);
```

Result: The pharmacy database was created with schema validation and sample data. Multi-collection aggregation queries were performed to generate order details, medicine sales, supplier revenue, low-stock medicines, and customer order summaries.